

A-level

ENVIRONMENTAL SCIENCE

Questions by topic: Global climate change - Set 1

Mark scheme

AQA spec 3.2.1.2; The physical environment; textbook pp. 89-107

Marking guidance

- Marking points are separated by bullets; each bullet is worth 1 mark unless stated.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	• Visible light passes through the atmosphere easily and is absorbed by the Earth's surface, which warms and emits infrared • Greenhouse gases (naturally water vapour and CO₂) absorb infrared better than the average for the atmosphere, converting it to heat and re-emitting some back to the surface • This keeps the surface about 33 °C warmer than it would otherwise be	3
01.2	• CO₂ is emitted in far larger quantities (every combustion process, deforestation) so its total effect is greatest • CO₂ has a long residence time so it accumulates; methane has a short residence time (12 years) so it is removed faster • CFC emissions are now very small following the Montreal Protocol	3
01.3	• Methane has a much higher global warming potential (28x) so each tonne removed has a bigger effect • Its short residence time (about 12 years) means atmospheric concentrations fall quickly once emissions are cut	2
01.4	<i>Any 2 from:</i> • Collect landfill / anaerobic digester gas and use as fuel • Changes in cattle diet / fewer cattle / less meat consumption • Draining rice paddies periodically • Anaerobic digestion of manure with gas capture	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 89-90; AQA spec 3.2.1.2.	
02.1	• $(7.7 - 3.9) / 7.7 \times 100$ • = 49.4% (accept 49%)	2

Question	Marking guidance	Marks
02.2	- Floating sea ice has already displaced its own mass of water; when it melts it contracts and occupies the same volume as the ice below the water line - Melting land ice (glaciers, ice sheets) adds new water to the sea, increasing its volume	2
02.3	- Ice and snow have a high albedo and reflect most sunlight; open sea water has a low albedo - As ice area shrinks more sunlight is absorbed by the sea and converted to heat, raising temperatures - This melts more ice: a positive feedback mechanism; thinner ice melts more easily each summer	3
02.4	- Thermal expansion: warmer sea water expands (slow because of the huge volume and high heat capacity; only surface water warms directly) - Melting of land ice (glaciers, Greenland and Antarctic ice sheets) adds water to the oceans	2
02.5	Ice shelves grounded on the seabed hold back / block the forward movement of the glaciers or ice sheet behind them; once removed the land ice flows faster (eg West Antarctica)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 94-96; AQA spec 3.2.1.2.	
03.1	- Trade winds blow westwards across the Pacific near the equator, driving a surface current in the same direction - Surface water moving away from South America is replaced by deep, cold water drawn upwards (upwelling) near the coast - The upwelling water is rich in nutrients (nitrates, phosphates) so large algal / phytoplankton blooms occur - These support a rich food web including the anchovy fishery; the water warms as it travels west	4
03.2	- The trade winds weaken or reverse so the surface current slows / reverses and warm water spreads east - Upwelling of nutrient-rich cold water off South America stops - Phytoplankton productivity collapses so the food web / anchovy population collapses; catch fell from about 7.7 to 1.7 million tonnes	3
03.3	<i>Any 2 from:</i> - Heavy rain and floods in the normally dry coastal deserts of Peru / South America - Drought and reduced rainfall in eastern Australia / Indonesia - Droughts in NE Africa, southern Africa, China - Fewer hurricanes in the North Atlantic / fewer cyclones in Japan	2
03.4	Events appear to be occurring more frequently and are driven by wind patterns that warming may alter, but the causes of ENSO are not fully understood and natural fluctuations make trends hard to separate	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 96-99; AQA spec 3.2.1.2, 3.2.2.3.	
04.1	- Friction with the prevailing south-westerly winds drags surface water north-eastwards from the tropical Atlantic - Water in the NE Atlantic cools, becomes denser (and more saline as ice forms) and sinks, drawing warm surface water in to replace it; the cold water returns southwards at depth (thermohaline circulation)	2

Question	Marking guidance	Marks
04.2	• Higher temperatures melt land ice on Greenland which flows into the North Atlantic • The fresh water dilutes the seawater, reducing its salinity and density • Less dense water sinks less easily so the flow rate of the current is reduced • Less heat is transported to NW Europe so the UK could become colder (like Moscow / Churchill at the same latitude) even while global temperatures rise	4
04.3	• Several interconnected systems act at once over different timescales: jet stream changes may raise temperatures, a slowing Conveyor may lower them, greenhouse gases raise them • Natural fluctuations (solar output, orbit) can hide or exaggerate anthropogenic change; individual events do not show trends • Time delays between cause and effect (oceans take a long time to warm because of volume and heat capacity); incomplete understanding of the processes	3
04.4	• Argo floats sink to a set depth for about 10 days, then surface and transmit temperature, salinity and position data; the sequence of positions shows current direction and speed (surface currents also by satellites and buoys)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 96-97; AQA spec 3.2.1.2, 3.2.2.3.	
05.1	• The climate naturally fluctuates (solar output, Earth's orbit) so extreme events can occur without human influence • An individual event, or even several, does not show a trend; the change may be in the frequency of events which can only be assessed years later • Changes occur at different time and spatial scales (local storm vs global trend); interconnected systems make attribution complex	3
05.2	• Towns have a heat-island effect (heat emissions, dark surfaces absorbing light) • As the town grows around the weather station the warming recorded is partly local, not global; older equipment / sparse global coverage also reduce reliability	2
05.3	• Tree rings: width of each annual ring shows the growth rate, indicating temperature / rainfall in that year; rings easily dated by counting; limitation: growth also affected by other factors (water, pests, shading), only covers the life of the tree / limited regions • Ice cores: annual snow layers compact into ice; air bubbles trapped in the ice give the CO₂ (and methane) concentration when trapped; oxygen isotope ratios indicate temperature; up to 800 000 years (Dome C); limitation: deep layers compressed / less distinct, dating needs radio-isotope analysis and volcanic ash layers; only polar regions • Others accepted: coral growth rings; pollen in lake sediments dated by radiocarbon	5
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 100-102; AQA spec 3.2.1.2.	

Question	Marking guidance	Marks
06.1	• LEO: high resolution and coverage of the whole Earth (including the poles) allows detailed mapping of ice area / thickness and forest cover • But each area is only revisited every 15 days so they cannot follow rapidly changing weather • Geostationary: constant view of the same region allows continuous monitoring of cloud / storm movements, though the poles are not visible and detail is lower	3
06.2	<i>Any 3 from:</i> • Wind velocity • Ocean currents / sea surface temperature • Wave height • Ice thickness • Vegetation cover • Sea level • Atmospheric temperature / composition	3
06.3	• Data for a past year (eg 1900) are fed in and the model's prediction for a later year (eg 2000) is compared with the real outcome • If the prediction matches, the model can be trusted (with caution); it is continually modified as more data are collected and differences analysed	2
06.4	• Models need data from all regions / oceans / altitudes; gaps (eg oceans, polar regions, past periods) introduce uncertainty into predictions • Unrepresentative data (eg urban heat islands, short time series) would bias the model; remote sensing helps fill gaps	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 102; AQA spec 3.2.1.2.	
07.1	• Negative feedback: a change causes other changes that reduce the rate or size of the original change, re-establishing equilibrium • Positive feedback: a change causes other changes that increase the original change	2
07.2	• Increased low-level cloud: higher temperatures increase evaporation and condensation, producing more cloud; clouds have a high albedo so more sunlight is reflected and warming is reduced • Increased photosynthesis: higher temperatures and CO₂ increase photosynthesis, removing CO₂; if stored in woody tissue atmospheric CO₂ rises less	4
07.3	• Permafrost: frozen waterlogged Arctic soils contain dead organic matter that decomposed anaerobically producing methane trapped in the frozen ground • Warming thaws the permafrost, releasing methane (a powerful greenhouse gas) and allowing further decomposition, causing more warming • Methane hydrate: methane from decomposed organic matter in deep sea sediments is held as a solid under high pressure and low temperature • If deep water warms the hydrate melts, releasing methane into the atmosphere and causing further warming	4

Question	Marking guidance	Marks
07.4	<i>Any 3 from:</i> • Reduced albedo as snow / ice melts • Ocean acidification reducing coral survival and carbon sequestration • Increased forest and peat fires releasing CO₂ • Faster decomposition of soil organic matter releasing CO₂ • Increased water vapour (a greenhouse gas) in warmer air • Increased cirrus (high) cloud which traps infrared	3
07.5	• A point at which natural processes triggered by human-caused warming become self-sustaining / unstoppable • So that stopping the original human activity would no longer stop climate change (eg runaway permafrost melt, methane hydrate release, forest fires)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 103-104, 371-372; AQA spec 3.2.1.2, 3.6.1.	
08.1	• Direct effect of abiotic change: species near the edge of their range of tolerance (temperature, water) may die out locally, eg sundews if marshes dry, frogs if climate drier; beech (shallow roots) suffers in drought more than oak • Changes in timing: earlier flowering / emergence may separate flowers from pollinators; plant toxins building up earlier may kill caterpillars; warmer winters disturb dormouse hibernation so they use fat reserves and starve • Biotic changes: food supply, predators, disease vectors (mosquitoes / malaria spreading), pollinators and seed dispersers may change • Distribution changes: species may colonise newly suitable areas (triton molluscs around the UK) but slow-growing species (trees) may disappear from old ranges faster than they colonise; suitable areas or biological corridors may not exist; human land use blocks movement • Population fragmentation: rising seas or upward retreat to mountains isolate populations; divided gene pools increase inbreeding; small populations may not be viable; no repopulation from elsewhere • Some species benefit (bats: warmer winters, more insects) but wetter stormier weather reduces feeding time	6
08.2	• All the species it depends on (food plants, pollinators, prey, mycorrhizae) will not move at the same speed • Less mobile species (plants, non-flying animals) cannot cross unsuitable areas without a biological corridor	2
08.3	• The temperature difference between mid-latitude and polar air drives the jet streams; a smaller difference makes the jet stream slower and more meandering (Rossby waves) • Weather systems stall over an area, giving prolonged drought or rainfall / floods; unusually cold or warm air carried north or south	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 90-92; AQA spec 3.2.1.2.	